Universität Bonn

Notes for the lecture

Combinatorics, Graphs, Matroids

held by

Dr. Ulrich Brenner

TEXed by

Daniel Emse

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Corrections and improvements

If you have corrections or improvements, contact me via (`daniel.emse@uni-bonn.de`).

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Lecture

Part I

Combinatorics

Chapter 1

Introduction

1.1 Fundamental Counting Rules

[08.10.2024, Lecture 1]

Goal: Count the number of elements in some finite sets.

Lemma 1.1: Fundamental counting rules

(*i*) *For pairwise disjoint and finite sets* $U_1, \dots, U_n$*, we have*

$$\left| \bigcup_{i=1}^{n} U_i \right| = \sum_{i=1}^{n} |U_i| . \tag{1.1}$$

(*ii*) *For finite sets* $U_1, \dots, U_n$*, we have*

$$|U_1 \times \cdots \times U_n| = \prod_{i=1}^{n} |U_i| . \tag{1.2}$$

(*iii*) *If there is a bijection between two sets* S *and* T*, then* $|S| = |T|$*.*

Notation. We call a set a n-set if it is a set with cardinality $n \in \mathbb{N}$ (where $0 \in \mathbb{N}$).

Proposition 1.2. *For two* n*-sets* A *and* B*, the number of bijections between* A *and* B *is* $n!$*.*

Proof. Let U the set of bijections from A to B. Apply induction in n:

$n = 0$: Trivial, because $0! = 1 = |U|$.

$n \mapsto n + 1$: Let $n > 0$ and let $x \in A$. For $i \in B$, let U_i be the set of all bijections $f : A \to B$ with $f(x) = i$. By induction, $|U_i| = (n-1)!$ for each $i \in B$. Then, $U = \bigcup_{i \in B} U_i$, so we get

$$|U| = \sum_{i \in B} |U_i| = n \cdot (n-1)! = n!. \tag{1.3}$$

□

Corollary 1.3. *The number of permutations of an* n*-set is* $n!$*.* □

Proposition 1.4. *The number of mappings from a* k*-set* A *to an* n*-set* B *is* n^k*.*

Proof. There is a bijection between the set of all mappings from A to B and the set $\underbrace{B \times \cdots \times B}_{k \text{ times}}$.

The number of these mappings is by Lemma 1.1.(*ii*) just $|B|^k = n^k$. □

Notation. We denote the set of all mappings from A to B by B^A.

Proposition 1.5. *The number of subsets of an n-set is 2^n.*

Proof. For an n-set A, there is a bijection between its power set and the mappings from A to $\{0,1\}$. Thus, the power set has 2^n elements. □

Notation.

(*i*) We denote the power set of a set A by 2^A.

(*ii*) The *binomial coefficient* $\binom{n}{k}$ is defined as the number of k-subsets of a n-set (for $k, n \in \mathbb{N}$).

What does *counting* mean? $\rightsquigarrow$ Possible answers:

(*i*) a direct, closed formula,

(*ii*) a sum,

(*iii*) a recursive formula

Example. Consider the number y_n of $0-1-2$-strings of length n with an even number of 1s and an odd number of 2s. Answers of types (*ii*) and (*iii*):

- We have
$$y_n = \sum_{i=0}^{\lfloor \frac{n-1}{2} \rfloor} \binom{2}{2i+1} 2^{2i}. \tag{1.4}$$
- We have
$$y_n = 3^{n-1} - y_{n-1} \quad (n > 0). \tag{1.5}$$

The proof of both formulas is exercise 1.2.

1.2 Elementary Counting Coefficients

Proposition 1.6. *For $n, k \in \mathbb{N}$ with $k \le n$, we have*

(*a*) $\binom{n}{k} = \binom{n}{n-k}$

(*b*) $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ *for* $k \ge 1$

(*c*) $\binom{n}{k} = \frac{n!}{(n-k)!k!}$.

Consider three kinds of rows in Pascal's triangle:

Row sums $\sum_{k=0}^{n} \binom{n}{k} = 2^n$ (both terms count the number of subsets of an n-set)

Column sums $\sum_{m=0}^{n} \binom{m}{k} = \binom{n+1}{k+1}$ for $n, k \in \mathbb{N}$ (where $\binom{m}{k}$ is the number of all $(k+1)$-subsets of $\{1, \ldots, n+1\}$ with the property that $m+1$ is the largest element in the subset)

Diagonal sums $\sum_{k=0}^{n} \binom{m+k}{k} = \binom{m+n+1}{n}$ for $m, n \in \mathbb{N}$ (for $k \in \{0, \ldots, n\}$, where $\binom{m+n}{k}$ is the number of n-subsets of $\{1, \ldots, m+n+1\}$ where $m+k+1$ is the largest number that is **not** contained in the subset)

Proposition 1.7. *For $x, y \in \mathbb{R}$, we have $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$.*

Proof. The coefficients of $x^k y^{n-k}$ in $(x+y)^n$ is $\binom{n}{k}$. □

Definition 1.8: Stirling number second kind: $S_{n,k}$

For $n, k \in \mathbb{N}$ let $S_{n,k}$ be the number of ways to partition an n-set in to k non-empty sets. The number $S_{n,k}$ is called *Stirling number of the second kind.*

Observation. For $n > 0$: $S_{n,0} = 0, S_{n,1} = n, S_{n,2} = 2^{n-1} - 1$.

Proposition 1.9. *For $n, k \in \mathbb{N} \setminus \{0\}$ we have $S_{n,k} = S_{n-1,k-1} + k \cdot S_{n-1,k}$.*

Proof. Let M be an n-set and $x \in M$. Then, the set of partitions of n into k subsets can be decomposed in the set of partitions with the set $\{x\}$ (there are $S_{n-1,k-1}$ of them) and the set of partitions where x is an element in a set with more than one element (there are $k \cdot S_{n-1,k}$ of them) □

[08.10.2024, Lecture 1]
[10.10.2024, Lecture 2]

Next goal: Count the number of mappings from an n-set to an r-set under certain constraints. To this end, we collect *counting coefficients*. We know already:

- $\binom{n}{k}$
- $S_{n,k}$: Number of ways to partition an n-set into k non-empty sets

Definition 1.10: $P_{n,k}$: Number of ways writing n as sum of k integers

For $n, k \in \mathbb{N}$, let $P_{n,k}$ be the number of ways to write n as the sum of k positive integers (without considering the order of the summands).

Example. $P_{6,3} = 3$, because the only possible partitions are: $6 = 4+1+1 = 3+2+1 = 2+2+2$.

Observation. For $n > 0$, we have $P_{n,0} = 0, P_{n,1} = 1, P_{n,2} = \lfloor \frac{n}{2} \rfloor$.

We can also consider *ordered partitions of numbers*:

Example. There are 6 ordered 3-partitions of the number 5: $5 = 3+1+1 = 1+3+1 = 1+1+3 = 2+2+1 = 2+1+2 = 1+2+2$.

Proposition 1.11. *The number of ordered partitions of a positive integer n into k summands is $\binom{n-1}{k-1}$.*

Proof. Write n as $n = \underbrace{1 + \cdots + 1}_{n \text{ times}}$. Any choice of $k-1$ of the $n-1$ „+“-signs gives you an ordered partition of n into k summands. Tis is a one-to-one correspondance. □

1.3 Assignments

Observations.

(*i*) The number of *surjective mappings* from an n-set N to an r-set R is $r! \cdot S_{n,r}$.

Proof. There are $S_{n,r}$ ways to partition n into r preimages and for each such partiton, we get $r!$ ways to assign the preimages to the elements of R. □

(*ii*) The number of *injective mappings* from an n-set N to an r-set R is

$$r(r-1)(r-2)\cdots(r-n+1). \tag{1.6}$$

Notation. For $n, r \in \mathbb{N}$, we call

$$r^{\underline{n}} := r \cdot (r-1) \cdot (r-2) \cdots (r-n+1) \tag{1.7}$$

the *n-th falling factorial* of r.

Analogously, the *n-th rising factorial* of r is

$$r^{\overline{n}} := r \cdot (r+1) \cdot (r+2) \cdots (r+n-1). \tag{1.8}$$

Definition 1.12: Multiset

A *k-multiset* over an n-set A is a set $B \subseteq A \times \mathbb{N}$ such that $\sum_{(a,i)\in B} i = k$ and $|\{(a,i) \in B \mid i \in \mathbb{N}\}| = 1$ for all $a \in A$.

$\rightsquigarrow$ Multisets can be considered as a set that may contain an element more than once. We write again $\{a_1, \ldots, a_k\}$ for a k-multiset (so $a_i = a_j$ is possible for $i \neq j$).

Proposition 1.13. *The number of k-multisets over an n-set is $\frac{n^{\overline{k}}}{k} = \binom{n+k-1}{k}$.*

Proof. There is a bijection f between the set of all k-multisets over $\{1, \ldots, n\}$ and the set of all k-subsets of $\{1, \ldots, n+k-1\}$. For a k-multiset $\{a_1, \ldots, a_k\}$ with $a_1 \leq a_2 \leq \cdots \leq a_{k-1} \leq a_k$, set $f(\{a_1, \ldots, a_j\}) = \{a_1, a_2+1, a_3+2, \ldots, a_k+(k-1)\}$. It is easy to see that this defines a bijection. □

Proposition 1.14. *For $n, r \in \mathbb{N}$, we have $r^n = \sum_{k=0}^{n} S_{n,k} r^{\underline{k}}$.*

Proof. For two sets A and B, let $\mathrm{Surj}(A,B)$ be the set of all surjective mappings from A to B. We know from Observation (i) that $|\mathrm{Surj}(A,B)| = |B|! \cdot S_{|A|,|B|}$. Let N be an n-set and R an r-set. Then, r^n is the number of mappings from N to R. We can partition

$$\begin{aligned} r^n &= \sum_{A \subseteq R} |\mathrm{Surj}(N,A)| = \sum_{k=0}^{r} \sum_{\substack{A \subseteq R \\ |A|=k}} |\mathrm{Surj}(N,A)| \\ &= \sum_{k=0}^{r} \binom{n}{k} k! S_{n,k} = \sum_{k=0}^{r} S_{n,k} r^{\underline{k}} = \sum_{k=0}^{n} S_{n,k} r^{\underline{k}} \end{aligned} \tag{1.9}$$

where the last step is possible because $S_{n,k} = 0$ for $k > n$ and $r^{\underline{k}} = 0$ for $k > r$. □

elements distinguishable		Mapping $f: N \to R$			
N	R	arbitrary	injective	surjective	bijective
Yes	Yes	r^n	$r^{\underline{n}}$	$r! S_{n,r}$	$\begin{matrix} 0 & (n \neq r) \\ n! & (n = r) \end{matrix}$
No	Yes	$\frac{r^{\overline{n}}}{n!}$	$\binom{r}{n}$	$\binom{n-1}{r-1}$	$\begin{matrix} 0 & (n \neq r) \\ 1 & (n = r) \end{matrix}$
Yes	No	$\sum_{k=1}^{r} S_{n,k}$	$\begin{matrix} 0 & (r < n) \\ 1 & (r \geq n) \end{matrix}$	$S_{n,r}$	$\begin{matrix} 0 & (n \neq r) \\ 1 & (n = r) \end{matrix}$
Yes	Yes	$\sum_{k=1}^{r} P_{n,k}$	$\begin{matrix} 0 & (r < n) \\ 1 & (r \geq n) \end{matrix}$	$P_{n,r}$	$\begin{matrix} 0 & (n \neq r) \\ 1 & (n = r) \end{matrix}$

Table 1.1: Number of mappings from an n-set N to an r-set R with and without additional constraints

First row completely covered.

Second row
- $\frac{r^{\overline{n}}}{n!}$ is the number of r-multisets of an n-set.
- $\binom{n-1}{r-1}$ is the number of ordered partitions of n into r summands.

1.4 Generalized Counting Coefficients

For $r \in \mathbb{C}$ and $k \in \mathbb{N}$, we define

(i) $r^{\underline{n}} := r \cdot (r-1) \cdot (r-2) \cdots (r-k+1)$ (with $r^{\underline{0}} = 1$)

(ii) $r^{\overline{n}} := r \cdot (r+1) \cdot (r+2) \cdots (r+k-1)$ (with $r^{\overline{0}} = 1$)

For $r \in \mathbb{C}, k \in \mathbb{Z}$, we define

$$\binom{r}{k} = \begin{cases} \frac{r^{\underline{k}}}{k!} & k \geq 0 \\ 0 & k < 0. \end{cases} \tag{1.10}$$

Proposition 1.15. *For $r \in \mathbb{C}, k \in \mathbb{Z}$, we have*

$$\binom{r}{k} = \binom{r-1}{k-1} + \binom{r-1}{k}. \tag{1.11}$$

Proof. If $k \leq 0$: trivial.

For $k > 0$, both sides of the equation are (for a fixed k) polynomials in r of degree k. They are equal for any $r \in \mathbb{N}$, so for an infinite number of values of r. $\rightsquigarrow$ Both polynomials are identical. □

[10.10.2024, Lecture 2]
[15.10.2024, Lecture 3]

Appendix

List of definitions

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